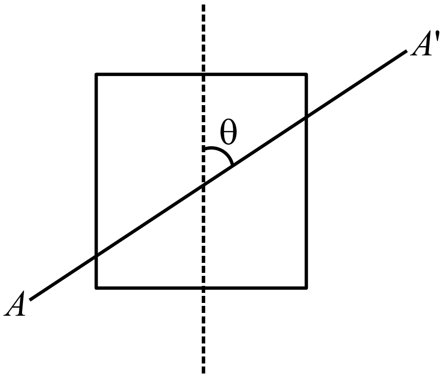
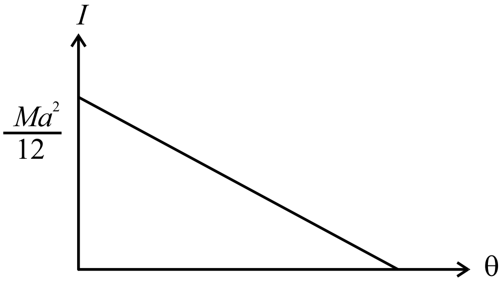
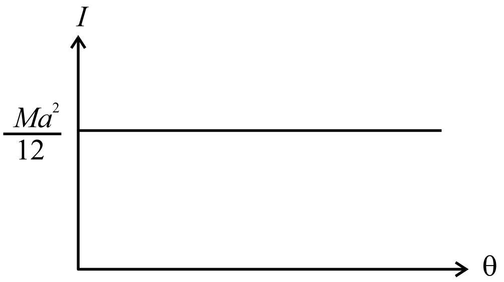
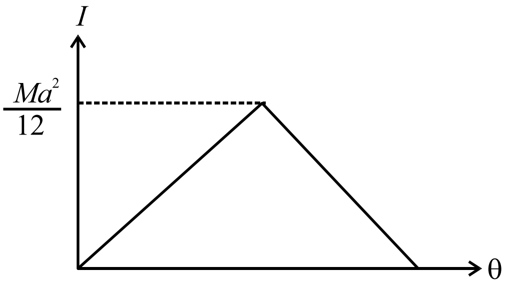
1. A uniform square sheet of mass 'M' and side 'a' rotates about an axis AA' that lies in the plane of the sheet and passes through its center of mass. The axis makes an angle θ with a vertical line of symmetry as shown.

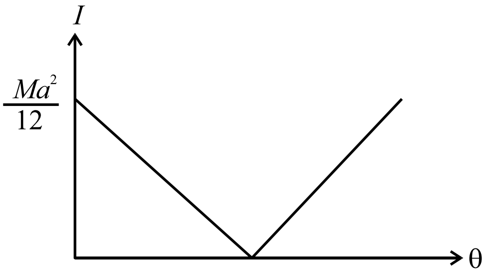


If the angle θ is continuously increased, which graph best represents the relationship between the moment of inertia (I) about this axis and the angle θ?

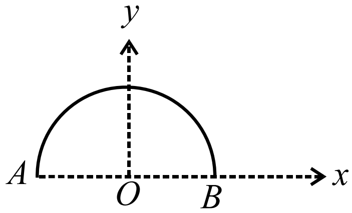
(1)


(2)


(3)


(4)


2. A thin wire of length 1 m is bent to form a semicircle as shown in the figure. The magnitude of the gravitational field at the center of the semicircle (point O) is $4\pi G$ N/kg, where G is the universal gravitational constant. What is the mass of the wire?

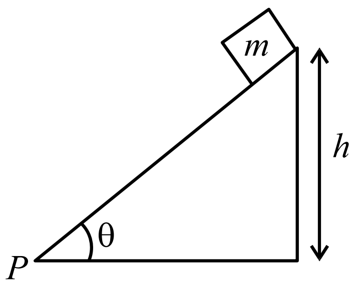


(1) 1 kg (2) 2 kg
(3) 4 kg (4) π kg

3. During a tensile strength test of a steel cable, a student increases the load such that the stress on the cable doubles. Assuming the test is conducted within the material's elastic limit, what is the effect on the Young's Modulus of the steel?

(1) It doubles
(2) It remains constant
(3) It is halved
(4) It quadruples

4. A small object of mass m is placed at the top of a fixed inclined plane of inclination θ. The object is at a vertical height h above the ground as shown. What is the magnitude of the torque produced by the object's weight about the vertex P? (neglect object size) (g is acceleration due to gravity)



(1) mgh
(2) $mgh\ \sin(\theta)$
(3) $mgh\ \tan(\theta)$
(4) $mgh\ \cot(\theta)$

5. Three point masses are placed in the x-y plane. A mass $m_1 = 2$ kg is at the origin $(0,0)$ m, a mass $m_2 = 9$ kg is at $(3,0)$ m, and a mass $m_3 = 32$ kg is at $(0,4)$ m. Masses m_2 and m_3 are fixed in place, while mass m_1 is free to move. What is the initial acceleration of mass m_1 due to the gravitational attraction from the other two masses? (Assume G is the universal gravitational constant in SI units).

(1) $G(\hat{i}+2\hat{j})$ m/s^2

(2) $G(2\hat{i}+4\hat{j})$ m/s^2

(3) $G(\hat{i}-2\hat{j})$ m/s^2

(4) $2G(\hat{i}+\hat{j})$ m/s^2

6. A solid sphere has mass M and radius R. A small test mass m is placed at a distance $3R$ from the center. What is the magnitude of the gravitational force exerted by the sphere on the test mass m?

(1) $\frac{GMm}{R^2}$ (2) $\frac{GMm}{2R^2}$

(3) $\frac{GMm}{9R^2}$ (4) Zero

7. A mechanical arm is acted upon by a force vector $F = (\hat{i}+2\hat{j}-3\hat{k})$N at a point with position vector $r_1 = (3\hat{i}+\hat{j}+2\hat{k})$m. What is the torque produced by this force about a pivot point located at $r_2 = (2\hat{i}-\hat{j}+\hat{k})$m?

(1) $(8\hat{i}-4\hat{j})$Nm

(2) $(-8\hat{i}+4\hat{j})$Nm

(3) $(-4\hat{i}-4\hat{j}+2\hat{k})$Nm

(4) $(-8\hat{i}-4\hat{j})$Nm

8. Young’s modulus (in N/m^2) for a perfectly rigid body is:

(1) Zero (2) Infinite

(3) 0.5 (4) 1

9. A newly discovered planet, "Kepler-X," has a mass M_p and a radius R_p. An object of mass m is dropped near its surface. The acceleration due to gravity it experiences is g_p. If the planet's mass were doubled ($2M_p$) and its radius were halved ($R_p/2$), what would be the new acceleration due to gravity, g_{new}?

(1) $2\,g_p$ (2) $4\,g_p$

(3) $8\,g_p$ (4) $\frac{g_p}{2}$

10. Two wires, P and Q, are made of the same alloy and have the same volume. Wire P has a cross-sectional area A and wire Q has a cross-sectional area $4A$. If a force F is required to stretch wire Q by an amount Δl, what force is needed to stretch wire P by the same amount Δl?

(1) $16F$

(2) $4F$

(3) $\frac{F}{4}$

(4) $\frac{F}{16}$

11. A solid circular disc of mass 50 kg and radius 1 m is rotating about its central axis perpendicular to the plane of the disc with angular speed 4 rad/s. The absolute value of external work done required on the disc to stop it is:

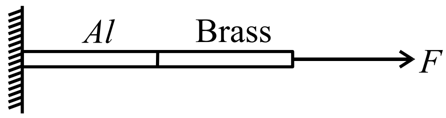
(1) 200 J

(2) 400 J

(3) 600 J

(4) 800 J

12. An aluminum wire and a brass wire of the same length are connected end-to-end. This composite wire is stretched by a force F as shown. If both wires have the same cross-sectional area, what can be said about the stress and strain in each wire? (Young's Modulus for brass is greater than for aluminum: $Y_{brass} > Y_{alum}$).



(1) Same stress, same strain.

(2) Same stress, different strain.

(3) Different stress, same strain.

(4) Different stress, different strain.

13. A body moves in a circle with angular acceleration $\alpha = 6t^2 - 2t$ (where t is in second and α is in rad/s^2). At t = 0, the ball has angular velocity of 10 rad/s. What is the angular velocity of body at t = 5 s?

(1) 235 rad/s

(2) 225 rad/s

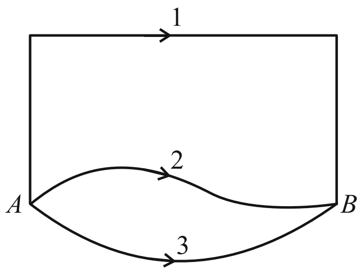
(3) 125 rad/s

(4) 50 rad/s

14. Elastic forces:
(1) Are always conservative
(2) Are not always conservative
(3) Are never conservative
(4) None of these

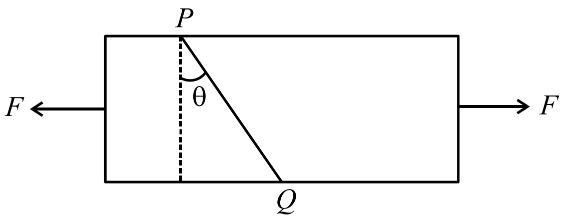
15. An object of mass m is lifted from the surface of the Earth (radius R) to an altitude of $h = \frac{R}{2}$. What is the change in its gravitational potential energy? (g is acceleration due to gravity)
(1) $\frac{1}{3}mgR$ (2) $\frac{1}{2}mgR$
(3) mgR (4) $\frac{2}{3}mgR$

16. A gravitational field is present in a region where a mass is shifted from A to B through different paths as shown. If W_1, W_2 and W_3 represent the work done by the gravitational force along the respective paths, then:



(1) $W_1 < W_2 < W_3$
(2) $W_1 = W_2 = W_3$
(3) $W_1 > W_2 > W_3$
(4) $W_1 > W_3 > W_2$

17. A rectangular block is subjected to a uniaxial tensile force F along its axis. An engineer needs to determine the plane PQ where the shear stress is at its absolute maximum. At what angle θ shown in figure, shear stress is maximum?



(1) $\frac{\pi}{4}$ (2) $\frac{\pi}{3}$
(3) $\frac{\pi}{6}$ (4) $\frac{\pi}{2}$

18. The axis X and Z in the plane of a planar body are mutually perpendicular and Y-axis is perpendicular to its plane. If the moment of inertia of the body about X and Y axes is respectively 30 kg m^2 and 40 kg m^2 then $M.I.$ about Z-axis in kg m^2 will be:
(1) 70 (2) 50
(3) 10 (4) Zero

19. A circular ring of wire of mass M and radius R is making n revolutions/sec about an axis passing through a point on its rim and perpendicular to its plane. The kinetic energy of rotation of the ring is given by:
(1) $4\pi^2MR^2n^2$ (2) $2\pi^2MR^2n^2$
(3) $\pi^2MR^2n^2$ (4) $8\pi^2MR^2n^2$

20. A particle of mass 30 g experiences a gravitational force of 6.0 N along positive x-direction. The gravitational field at that point is:
(1) $-200\,\hat{j}$ N/kg
(2) $200\,\hat{j}$ N/kg
(3) $-200\,\hat{i}$ N/kg
(4) $200\,\hat{i}$ N/kg

21. A thin, uniform spherical shell has a radius of 7 cm. The magnitude of the gravitational field at a distance of 10 cm from its center is 8 N/kg. What is the magnitude of the gravitational field at a distance of 15 cm from its center?
(1) 18 N/kg
(2) 12 N/kg
(3) $\frac{32}{9}$ N/kg
(4) $\frac{16}{3}$ N/kg

22. In an arrangement four particles, each of mass 2 kg are situated at the coordinate points (3, 2, 0) m, (1, –1, 0) m, (0, 0, 0) m and (–1, 1, 0) m. The moment of inertia (in kg-m^2) of this arrangement about the Z-axis will be:
(1) 8
(2) 16
(3) 43
(4) 34

[Contd...

23. A rotating table completes one rotation in 10 sec and its moment of inertia is 100 kg-m^2. A person of 50 kg mass stands at the centre of the rotating table. If the person moves 2 m from the centre, the angular velocity of the rotating table in rad/sec. will become:

(1) $\frac{2\pi}{30}$

(2) $\frac{20\pi}{30}$

(3) $\frac{4\pi}{3}$

(4) 2π

24. Copper of fixed volume V is drawn into wire of length l. When this wire is subjected to a constant force F, the extension produced in the wire is Δl. Which of the following graphs is a straight line?

(1) Δl versus $\frac{1}{l}$

(2) Δl versus l^2

(3) Δl versus $\frac{1}{l^2}$

(4) Δl versus l

25. The rotational kinetic energy of a rigid body of moment of inertia 5 kg-m^2 is 10 joules. The angular momentum about the axis of rotation would be:

(1) 100 joule-sec

(2) 50 joule-sec

(3) 10 joule-sec

(4) 2 joule -sec

26. Two point objects of mass $2x$ and $3x$ are separated by a distance r. Keeping the distance fixed, how much mass should be transferred from $3x$ to $2x$, so that gravitational force between them becomes maximum?

(1) $\frac{x}{4}$

(2) $\frac{x}{3}$

(3) $\frac{x}{2}$

(4) $\frac{2x}{3}$

27. A block of mass 'm' hangs from a uniform wire of mass 'M', area of cross-section A length L and Young's modulus Y. The total elongation in the wire is: (g is acceleration due to gravity)

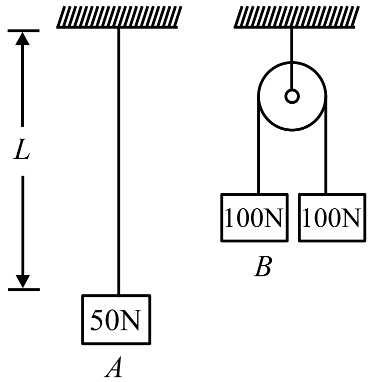
(1) $\frac{mgL}{AY}$

(2) $\frac{mgL}{2AY}$

(3) $\frac{mgL}{AY}+\frac{MgL}{2AY}$

(4) $\frac{mgL}{2AY}+\frac{MgL}{AY}$

28. In an experiment (Case A), a steel wire of length L is suspended from the ceiling. When a block weighing 50 N is attached to its free end, the wire is observed to elongate by 2.0 mm. In a second experiment (Case B), the same wire of the same length is passed over a frictionless pulley, and blocks of 100 N are suspended on both sides, as shown. What will be the elongation of the wire in Case B?



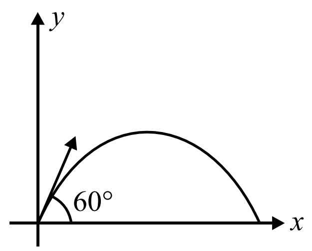
(1) 1.0 mm

(2) 2.0 mm

(3) 4.0 mm

(4) 8.0 mm

29. A particle of mass 2 kg is projected at an angle of 60° above the horizontal, with a speed 20 m/s. The magnitude of angular momentum of the particle about point of projection, when it is at maximum height will be: (g = 10 m/s^2)



(1) 300 kg m^2/s

(2) $120\sqrt{3}$ kg m^2/s

(3) $40\sqrt{3}$ kg m^2/s

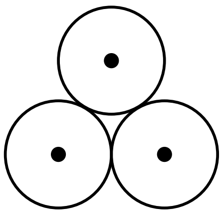
(4) 150 kg m^2/s

30. Two point masses, one of mass *m* and the other of mass 2*m*, are fixed at a separation distance of *r*. Consider an axis of rotation that is perpendicular to the line segment connecting the two masses. What is the minimum possible moment of inertia this system can have with respect to such an axis?

(1) $\left(\frac{1}{2}\right)mr^2$ (2) $\left(\frac{2}{3}\right)mr^2$

(3) $\left(\frac{3}{2}\right)mr^2$ (4) $2\ mr^2$

31. Three solid spheres of mass *M* and radius *R* are placed in contact as shown in figure. Find the gravitational potential energy due to interaction of masses in the system?



(1) $-\frac{3}{2}\frac{GM^2}{R}$ (2) $\frac{3}{2}\frac{GM^2}{R}$

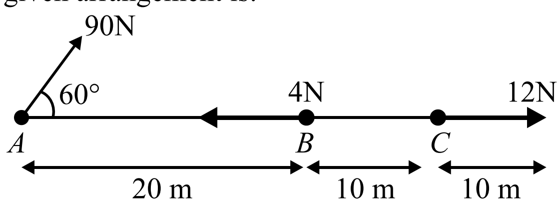
(3) $\frac{5}{2}\frac{GM^2}{R}$ (4) $-\frac{5}{2}\frac{GM^2}{R}$

32. A ladder rests against a frictionless vertical wall, with its upper end 6 m above the ground and the lower end 4 m away from the wall. The weight of the ladder is 500 N and its *C.G.* at 1/3rd distance from the lower end along the ladder's length. Wall's reaction will be closest to, (in Newton)

(1) 111 (2) 333

(3) 222 (4) 129

33. Net torque about point *A* by all the forces in the given arrangement is:



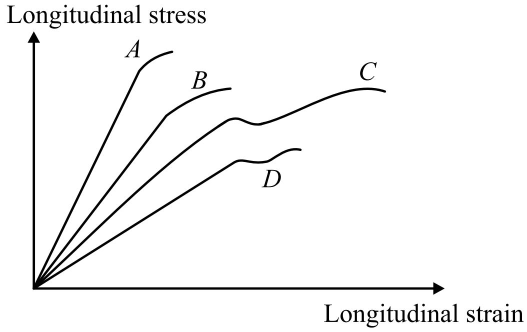
(1) 120 N-m

(2) 80 N-m

(3) 90 N-m

(4) 0 N-m

34. Longitudinal stress v/s longitudinal strain curve for four different materials *A*, *B*, *C* and *D* is shown. Arrange young's modulus of materials in increasing order.



(1) $C < B < A < D$ (2) $D < C < B < A$

(3) $B < C < A < D$ (4) $A < B < C < D$

35. A string is wrapped around the rim of a wheel of moment of inertia 0.40 kg m^2 and radius 10 cm. The wheel is free to rotate about its axis. Initially the wheel is at rest. The string is now pulled by a constant force *F*. The angular velocity of the wheel after 10 s is 100 rad/s, the value of *F* is:

(1) 50 N (2) 40 N

(3) 120 N (4) 150 N

36. If the angular speed of earth becomes twice of its present value, then the weight of the person standing on north pole will be: (consider rotational effect of earth only)

(1) halved (2) doubled

(3) remain same (4) quadrupled

37. Consider the following statements:

(a) In straight-line motion with constant velocity, the angular momentum of a particle about any point in space remains constant.

(b) A couple (two equal and opposite forces, separated by some distance) always produces the same torque about any point in the plane of the forces.

(c) Angular momentum of an isolated system is conserved only when net external torque on the system is zero.

(d) In the perpendicular axis theorem, the axis must pass through the centre of mass of the body, otherwise the theorem cannot be applied.

Which of the above statements are **correct**?

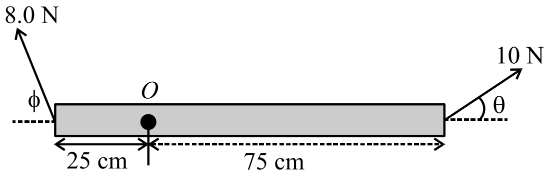
(1) (a), (b) and (c) only

(2) (a) and (c) only

(3) (b), (c) and (d) only

(4) (a), (c) and (d) only

38. The image shows a massless rod pivoted at the point O. A force of 8.0 N is applied at the left end, and a force of 10 N is applied at the right end, at the angles shown. If the rod is in rotational equilibrium, what is the relationship between the angles φ and θ?



(1) $\sin(\varphi) = 3.75 \sin(\theta)$
(2) $\sin(\theta) = 3.75 \sin(\varphi)$
(3) $\cos(\varphi) = 3.0 \sin(\theta)$
(4) $\sin(\varphi) = 3.0 \sin(\theta)$

39. A wire 8m in length suspended vertically stretches by 20 mm when mass of 3 kg is attached to the lower end. The elastic potential energy gain by the wire is: (take $g = 10$ m/s^2);

(1) 0.3 J (2) 3 J
(3) 30 J (4) 300 J

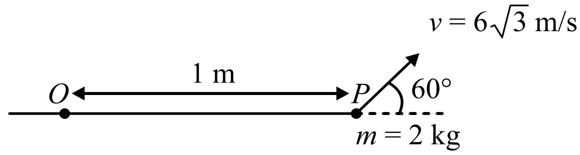
40. Choose the **correct** statement/s.
Statement I- Gravitational force is independent of the medium, temperature and pressure.
Statement II- Gravitational force is conservative force.
Statement III- Gravitational force is a weak force. This force is significant mainly in large size heavy bodies.
Statement IV- Gravitational force may be repulsive in some cases.

(1) I, II and IV (2) I, II and III
(3) I, II, III and IV (4) II, III and IV

41. Given below are two statements: One is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Bulk modulus of incompressible liquid is zero.
Reason R: The volume of incompressible liquid does not change by applying force.
In the light of the above statements, choose the **correct** answer from the options given below:

(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is NOT the correct explanation of A.

42. The magnitude of angular momentum of a particle of mass 2 kg about point O at the instant as shown in the figure, is:



(1) 6 kgm^{2}s^{-1}
(2) 9 kgm^{2}s^{-1}
(3) 18 kgm^{2}s^{-1}
(4) $9\sqrt{3}$ kgm^{2}s^{-1}

43. If the relative change in each edge of cube is 0.01, the volumetric strain is:

(1) 0.01
(2) 0.02
(3) 0.04
(4) 0.03

44. Two uniform identical discs rotating clockwise with angular velocities ω and 2ω separately about their axis of rotation passing through their centres are brought in contact. If the final angular velocity of the combined system is ω_f, then $\frac{\omega_f}{\omega}$ is:

(1) 1.5
(2) 2
(3) 0.5
(4) 4

45. Mass of thin uniform circular ring is M and its radius is R. Find the magnitude of maximum gravitational field on its axis;

(1) $\frac{2}{3\sqrt{3}}\frac{GM}{R^2}$
(2) $\frac{GM}{R^2}$
(3) $\frac{2GM}{R^2}$
(4) $\frac{3}{2\sqrt{2}}\frac{GM}{R^2}$